

DATABASE MANAGEMENT

UNIT-1

DBMS – A Historical Perspective

1. In early days, data was stored manually in paper files.
2. Organizations used file-based systems before databases.
3. File systems were the first method of storing computer data.
4. Data was stored in separate files for each application.
5. Each program had its own data file.
6. There was no sharing of data between applications.
7. This led to data redundancy (same data repeated).
8. Data inconsistency was a major problem.
9. Updating data in one file did not update others.
10. File systems were difficult to manage.
11. Searching data in large files was slow.
12. Data security was very weak in file systems.
13. No centralized control over data.
14. Data duplication increased storage cost.
15. Program-data dependency existed in file systems.
16. Any change in data format required program changes.
17. This created maintenance problems.
18. To solve these issues, database systems were introduced.
19. The concept of DBMS started in the 1960s.
20. Early databases were hierarchical databases.
21. IBM developed the first hierarchical model (IMS system).
22. In hierarchical model, data is arranged in tree structure.
23. Later, network database model was introduced.
24. CODASYL model supported network databases.

25. Network model allowed many-to-many relationships.
26. Both hierarchical and network models were complex.
27. In 1970, E.F. Codd introduced relational model.
28. Relational model is based on tables (relations).
29. It simplified data storage and retrieval.
30. SQL language was developed for relational databases.
31. Relational DBMS became very popular.
32. It reduced data redundancy.
33. It improved data consistency.
34. It supported data independence.
35. IBM developed System R (early relational DBMS).
36. Oracle was one of the first commercial RDBMS.
37. MySQL and PostgreSQL came later as open-source systems.
38. In 1980s, object-oriented databases were introduced.
39. They supported complex data types like images and videos.
40. Object-relational databases combined both models.
41. In 1990s, data warehouses were introduced.
42. Data mining became popular for analysis.
43. DBMS started supporting large-scale enterprise systems.
44. Internet increased demand for databases.
45. Web applications used backend databases.
46. Cloud databases were introduced later.
47. Examples: AWS RDS, Google Cloud SQL.
48. NoSQL databases came for big data systems.
49. Examples of NoSQL: MongoDB, Cassandra.
50. Today DBMS supports AI, cloud, and big data applications.